

Clinical Guideline for: **Diabetes in pregnancy management** Northern Maternity Services

SUMMARY

This guideline outlines the management of pregnant women with pre-existing diabetes (type 1 and 2) and gestational diabetes from pre-conception to post-natal period.

KEY POINTS

This guideline is updated according to the recommendations from the National Pregnancy in Diabetes Audit 2021 and 2022, England and Wales, incorporated into Saving Babies Lives: care Bundle Version 3.

Two main aims of introducing management are:

- To set up the standard care pathways for MDT management of these women throughout pregnancy, with increased access to 'joined-up' support for their complex care needs.
- To improve glucose control during pregnancy by supporting high-risk women not achieving safe pregnancy glycaemic targets and by high uptake of digital glucose monitoring technology.

The essential elements of this guideline are:

- Care for pregnant women with Type 1 and 2 Diabetes is provided by a multidisciplinary team at a dedicated one-stop clinic.
- Women with type 1 diabetes should be offered real time continuous glucose monitoring (CGM) and be provided with appropriate education and support to use this.
- Measurement of HbA1C for all women with T1D and T2D in the third trimester.
- All women must have a digital blood sugar record in their maternity documents.
- Women with type 1 and 2 diabetes and HbA1C ≥ 48 to have a formal MDT to review ongoing care.
- Women with active retinopathy and diabetic nephropathy/chronic kidney disease should be referred to the regional Maternal Medicine Centre (southwest Maternal Medicine Network) for discussion and further management.
- **To implement an agreed pathway for women presenting with Diabetic Ketoacidosis.**

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1. INTRODUCTION

- 1.1 This guideline covers managing diabetes and its complications in a woman who is planning a pregnancy or is already pregnant by evidence-based NICE (December 2020) guidelines and the Joint British Diabetes Society (JBDS) guidance No: 12, 2022.
- 1.2 The guideline is further updated following the publication of Saving Babies Lives: Care Bundle version 3 (2023) and the findings of the National Pregnancy in Diabetes (NPID) audit (2021-2022), England and Wales.

2. BACKGROUND

- 2.1. Poor control of diabetes before conception and during pregnancy increases perinatal morbidity and mortality as well as serious maternal complications. In England and Wales, there has been persistently high perinatal mortality with no improvement over the past five years.
- 2.2. A recent NPID audit identified rising rates for serious maternal outcomes with T2D and their babies in 2022 (5.5% in 2014-2020, 4.9% in 2021, and 6.6% in 2022).
- 2.3. Similarly, there has been a significant increase in the percentage of pregnancies with congenital anomalies and of pregnancies with other serious outcomes (excluding congenital anomalies) in T2D, compared to T1D. Serious outcomes include miscarriage, stillbirth, neonatal death, or birth defects.
- 2.4. The latest MBBRACE report highlighted the very high risk of fetal death (SB 160/1000 births) associated with Diabetic Ketoacidosis.

3. ABBREVIATIONS/ DEFINITIONS

T1D - Type 1 Diabetes
 DSN - Diabetic specialist nurse
 T2D - Type 2 Diabetes
 CMW - Community midwife
 GDM - Gestational Diabetes Mellitus.
 JBDS - Joint British Diabetes Society
 CSII - Continuous subcutaneous insulin infusion
 rtCGM - real-time Continuous glucose monitor
 isCGM - intermittently scanned Continuous glucose monitor
 VRIII - Variable rate intravenous insulin infusion
 CBG - Capillary Blood glucose
 HbA1C - Glycated haemoglobin
 DKA - Diabetic ketoacidosis
 SC - Subcutaneous
 ACR - Albumin creatinine ratio
 OGTT - Oral glucose tolerance test
 eGFR - Estimated glomerular filtration rate
 BMI - Body mass index
 BP - Blood pressure
 PET - pre-eclampsia

4. GENERAL PRINCIPLES OF MANAGING WOMEN WITH DIABETES

4.1 Pre-conception care

- Refer women with pre-existing diabetes who are planning pregnancy to the pre-conception clinic, NDDH, by any health care professional. Pre-conception counselling is vital because reasonable glycaemic control before conception (monthly HbA1c below 48 mmol/mol) is shown to reduce the risk of miscarriage, congenital malformation and other severe fetal and maternal risks. Women with diabetes whose HbA1c is above 86mmol/mol should be strongly advised to avoid pregnancy and to use contraception.
- At the pre-conception clinic:
 - a. Stop angiotensin-converting enzyme inhibitors and angiotensin-II receptor antagonists before conception or as soon as pregnancy is confirmed. Similarly, stop statins before pregnancy because of their teratogenic effects. Use alternative antihypertensive agents that are suitable for pregnant women.
 - b. When women with diabetes who have already been taking insulin or oral hypoglycaemic agents need adjusting or changing them, diabetic team should be involved.
 - c. Recommend retinal assessment to women with diabetes seeking pre-conception (unless an annual retinal assessment has occurred within the previous six months).
 - d. Recommend renal assessment, with a measure of microalbuminuria and consider referral to a nephrologist if serum creatinine is 120 micromole/litre or more or the eGFR is less than 45 ml/minute/1.73 m² or the albumin: creatinine ratio is greater than 20mg/mmol.
 - e. Advise women to take folic acid (5 mg/day) until 12 weeks of gestation to reduce the risk of having a baby with a neural tube defect.

4.2. Antenatal Management

4.2.1. Antenatal appointments

- Antenatal care for women with pre-existing diabetes (T1D and T2D) will be provided at the MDT one-stop clinic (maternal diabetic clinic) by a team consisting of an Obstetric consultant, Diabetes consultant, Diabetes Specialist Nurse and Diabetes dietician. Early booking (by ten weeks) appointment is advised for women with pre-existing diabetes.
- Prescribe Aspirin 150 mg daily from 12 weeks of pregnancy until delivery in those with T1D and T2D to reduce the risks of severe hypertensive disorders in pregnancy.
- Aim of the review are:
 - a. To monitor treatment during pregnancy.
 - b. To discuss, agree and document a peri-partum diabetes management plan.
 - c. To explain implications of poor blood glucose control during pregnancy and labour: increased risk of having a macrosomic baby with subsequent birth trauma (shoulder dystocia, perineal tear, and post-partum haemorrhage), in-utero fetal death, induction of labour and caesarean section, neonatal hypoglycaemia, and perinatal death.

- d. To monitor fetal growth and well-being. A fetal echocardiogram is offered to all women with T1D and those with poorly controlled T2D at 18-20+6 wks., followed by ultrasound monitoring of fetal growth and amniotic fluid volume every four weeks from 26 weeks. More frequent fetal monitoring may be required for women with diabetes and other risks of intrauterine growth restriction (including macrovascular disease and nephropathy).
- Minimum documentation should be recorded, including a) the type, frequency of glucose monitoring and target BG, b) if CGM, TIR above, below and in the target, c) HbA1C levels at booking and third trimester, d) if insulin treatment, method of delivery and types of insulin.

4.2.2. Screening and diagnosis for GDM

- A community midwife should offer women with the following risk factors when booking a visit screening for gestational diabetes:
 - BMI above 30 kg/m²
 - Previous macrosomic baby (4.5 kg or above (OR) $\geq$ 95th CC at birth)
 - Previous shoulder dystocia
 - Previous unexplained stillbirth
 - Previous gestational diabetes
 - First-degree relative with diabetes
 - Family origin with a high prevalence of diabetes: South Asian (India, Pakistan or Bangladesh), Black Caribbean and Middle Eastern (Saudi Arabia, United Arab Emirates, Iraq, Jordan, Syria, Oman, Qatar, Kuwait, Lebanon or Egypt).
 - Recent steroid therapy or antipsychotic medications
- Consider screening for gestational diabetes in the following conditions until 34 weeks gestation by OGTT and three days- capillary blood glucose monitoring thereafter, provided the result would be expected to change the future management.
 - Polyhydramnios (deepest pool > 8cm)
 - Glycosuria of 2+ or above on one occasion or 1+ or above on more than one occasion
 - Accelerated fetal growth is suspected on the growth scan.
- Book 2-hour 75 g OGTT for screening of gestational diabetes in women with risk factors at 24-28 weeks by a CMW. Do not use fasting plasma glucose, random blood glucose, urinalysis for glucose or HbA1c for screening purposes.
- For women who have had gestational diabetes in a previous pregnancy: Either early self-monitoring of blood glucose or a 75 g 2hour OGTT as soon as possible after booking (whether in the first or second trimester), and a further 75 g 2-hour OGTT at 24–28 weeks if the results of the first OGTT are normal.
- The midwife requesting the OGTT takes responsibility for checking the result and informing the DSN team (x 2726) if it is abnormal. A colleague should delegate this responsibility if the requester is on leave. Arrangements should be made for the woman to be seen by the diabetes team within one week. Dietary advice and glucose monitoring will be commenced ASAP before a clinic and scan appointment can be made.

4.2.3. Monitoring blood glucose and ketones during pregnancy

a. Glucose monitoring method

- Offer rtCGM to all women with T1D.
- Consider rtCGM for pregnant women who are not T1D but on insulin therapy if they have severe hypoglycaemia (with or without impaired awareness) or unstable blood glucose levels despite efforts to optimise glycaemic control.
- Objective recording of BG levels should be documented in the patient's electronic records.

b. Self- monitoring

- Agree on individualised targets for self-monitoring blood glucose with women, considering the risk of hypoglycaemia. If it is safely achievable, women with diabetes should aim for a target blood glucose of the following levels:
 - 5.3mmol/l fasting
 - 7.8mmol/l 1 hour after meals
- Women with T1D, T2D and Gestational diabetes who are on a multiple daily insulin injection regimen should be advised to test fasting, pre-meal, 1-hour post-meal and bedtime blood glucose levels daily.
- Advise pregnant women with type 2 diabetes or gestational diabetes to test their fasting and 1-hour post-meal blood glucose levels daily if they are:
 - managing their diabetes with diet and exercise changes alone **or**
 - taking oral therapy or single-dose intermediate-acting or long-acting insulin.
- Women with continuous glucose monitoring should have an individualised target, as agreed by diabetic team.

c. Ketones monitoring

- Supply pregnant women with Type 1 diabetes, blood ketone testing strips and a meter to test for ketonaemia. Advise all pregnant women with diabetes to seek urgent medical advice if they become hyperglycaemic or unwell.

d. HbA1C monitoring

- Measure HbA1c at booking and the beginning of 3rd trimester (28/40) to both pre-existing diabetes and GDM to assess the risks for the pregnancy and modify the treatment.

e. Action plan if the glycaemic targets are not achieved.

- If HbA1C >48 mmol/mol at 28/40, formal MDT should be considered for on-going care.
 - a) further support from diabetic SN or dietician
 - b) alternative monitoring or treatment
 - c) increased fetal surveillance
 - d) repeat discussion of fetal and neonatal risks

4.2.4. Hypoglycaemic agents - GDM

- **Trial of diet and exercise** to all women with a fasting blood glucose of $<7.0\text{mmol/l}$. If fail to maintain blood glucose targets during 1–2 weeks or if ultrasound investigation suggests fetal macrosomia (abdominal circumference above the 70th percentile), **advise pharmacological treatment with metformin**.
- **Insulin treatment** if metformin is not tolerated or blood glucose targets are not being met.
- **Immediate treatment with insulin**, with or without metformin, should be offered to all women with fasting blood glucose value of $>7.0\text{mmol/mol}$ or above at diagnosis or between 6.0 and 6.9 mmol/l at diagnosis with foetal macrosomia or polyhydramnios on ultrasound.

4.2.5. Insulin treatment - pre-existing diabetes

- Use isophane insulin (NPH insulin) as the first choice for long-acting insulin during pregnancy. Consider continuing treatment with long-acting insulin analogues (insulin detemir or insulin glargine) for women with diabetes who have established reasonable blood glucose control before pregnancy. Rapid-acting insulin analogues (aspart and lispro) do not adversely affect the pregnancy or the baby's health.
- Offer continuous subcutaneous insulin infusion (CSII or insulin pump therapy) if multiple daily insulin injections do not obtain adequate glycaemic control without significant disabling hypoglycaemia.
- Advise women with insulin-treated diabetes of the risks of hypoglycaemia and impaired awareness of hypoglycaemia in pregnancy, particularly in the first trimester. Rapid-acting insulin analogues (aspart and lispro) have advantages over soluble human insulin during pregnancy and consider their use.
- Advise pregnant women with insulin-treated diabetes to always have a fast-acting form of glucose available (for example, dextrose tablets or glucose-containing drinks). Provide glucagon to pregnant women with type 1 diabetes for use if needed. Explain how to use it to the woman, her partner, or other family members.

4.2.6. Retinal assessment during pregnancy

- Advise retinal assessment as soon as possible after the first contact in pregnancy in women with pre-existing diabetes if it has not been performed in the preceding three months and again at 28 weeks if the initial evaluation is normal. If any diabetic retinopathy is present, ophthalmologist involvement is necessary for treatment or further follow-ups.
- Diabetic retinopathy should not be considered a contraindication to rapid optimisation of glycaemic control in women who present with a high HbA1c in early pregnancy.
- Women who have pre-proliferative diabetic retinopathy or any form of diabetic retinopathy diagnosed during pregnancy should have ophthalmological follow-up for at least six months following the birth of the baby.

4.2.7. Renal assessment during pregnancy

- Arrange renal function assessment at the first contact in pregnancy if it has not been undertaken in the preceding three months in women with pre-existing diabetes.
- Refer to a nephrologist if serum creatinine is 120 micromole/litre or urinary albumin: creatinine ratio exceeds 20mg/mmol or total protein excretion exceeds two g/day.
- Consider thromboprophylaxis for women with significant proteinuria.

Women with complex diabetes, retinopathy and/or kidney impairment requiring treatment during pregnancy can be discussed with the regional maternal medicine network

4.2.8. Diabetic Ketoacidosis (DKA)

a. Diagnosis

1. Presence of Diabetes mellitus (of any kind, DKA can occur in pregnancy in a woman with known diabetes with a normal blood glucose level) AND:
2. Ketosis: urinary ketones >++ or blood ketones >3.0 mmol/L (high risk 1.5 mmol/L) AND
3. Acidosis: blood gas pH <7.3 and/or bicarbonate <15 mmol/L (N.B. bicarbonate is reduced in pregnancy). Use venous blood gases.

b. Treatment

1. Involve the medical or diabetes team and anaesthetic team urgently.
2. DKA in pregnancy should be managed in HDU/ICU.
3. Treat as per separate trust guideline on DKA.

4.2.9. Management of diabetes during steroid use for women on diet control / oral treatment and /or single or multiple dose insulin therapy.

- Administration of antenatal steroids for fetal lung maturity is recommended before 34+0 weeks and considered among women at risk for preterm birth between 34+0 and 35+6 weeks. Deterioration of blood glucose levels is anticipated for 2 to 3 days and should be actively managed. JBDS recommends regular monitoring of BG levels in these women.
- Subcutaneous (SC) insulin may need to be started in women managed by diet or metformin. An increase in insulin dose, typically by 50%, is required for those already on insulin.
- If BG levels remain higher than the target on two consecutive occasions, VRIII may be used. If VR is used in this context, the following approach is suggested by JBDS:
 - Check U+Es before starting VRIII to monitor fluid balance and electrolyte abnormalities and repeat 24 hourly.
 - Appropriate documentation and education are needed to prevent insulin errors.
 - Start VRIII (50 units of human soluble insulin [Humulin® S or Actrapid®] made up to 50 mL with 0.9% NaCl) to achieve either the target blood glucose of 4.0 – 7.8 or 5.0 – 8.0 mmol/L. When on VRIII, check capillary blood glucose level hourly.
 - Continuous intravenous insulin may be needed until 12 hours after the administration of the second dose of steroids.

- Basal insulin needs to be continued as usual. Mealttime insulin is continued if the woman is eating and drinking to manage glycaemic excursions adequately after a meal.
- The recommended substrate fluid to be administered alongside the VRIII is 0.9% NaCl with 5% glucose and 0.15% KCl (20 mmol/L) at 50 mL/hr. Pure dextrose-containing fluids should be avoided due to the risk of maternal and neonatal hyponatraemia and hypokalaemia.
- Additional i.v. Fluids may be needed if the patient is not eating or drinking reliably.
- Fluids, particularly dextrose-containing fluids, may have to be restricted in patients who are at risk of or already have hyponatraemia.
- In some cases, insulin without substrate fluids may have to be used (e.g., difficult i.v. access, fluid overload states like toxæmia, hyponatraemia, or risk of hyponatraemia). Please consult senior medical/ obstetric and anaesthetic staff as needed.

4.3 Intrapartum care

a. Timing and mode of delivery

- Discuss the timing and mode of birth with the woman during the third trimester. The diabetic team should document a plan for the management of diabetes during labour or caesarean section at 36 weeks of gestation.
- Advise women with pre-existing diabetes and no other complications to have an elective induction of labour or elective caesarean section between 37+0 and 38+6 weeks of pregnancy. Consider elective birth before 37+0 weeks if there are any metabolic or other maternal or fetal complications.
- Advise women with gestational diabetes to give birth no later than 40+6 weeks and offer elective delivery to those who have not delivered by this time. Consider elective birth before 40+6 weeks if there are any maternal or fetal complications for those women with gestational diabetes.
- Explain the risks and benefits of vaginal birth, induction of labour and caesarean section to pregnant women with diabetes who have an ultrasound-diagnosed macrocosmic fetus (EFW>95th CC).

b. Intrapartum glycaemic control

- Continuous electronic fetal monitoring is recommended for all women with diabetes due to the increased perinatal mortality rate.
- The paediatrician and NICU should be informed of impending delivery.
- Monitor hourly blood glucose (capillary, Flash or CGM) once in established labour or after artificial membrane rupture or on admission for caesarean section. If general anaesthesia is used, monitoring should be every half an hour from induction of GA until the woman is fully conscious.
- Women may choose to remain on insulin pump therapy during labour and caesarean section.
- If taking oral hypoglycaemic agents, discontinue when in established labour or if nil by mouth.
- Aim to maintain glucose levels (capillary, Flash or CGM) during labour 5.0 – 8.0 mmol/L. (JBDS 2022).

- **VRIII (variable rate intravenous insulin infusion)** should be considered;
 - If two consecutive blood glucose levels are above the target range (8.0 mmol/L). The second blood glucose should be checked within an hour to prevent any delay in starting VRIII.
 - In some women with type 1 diabetes with erratic blood glucose control, it may be justified to start VRIII from the time of onset of labour, irrespective of BG readings, for more effective glucose control.
 - Continue meal-time insulin when VRIII is started if the woman is eating and drinking.
 - CGM should NOT BE USED to guide VRIII doses. Always use capillary blood glucose to adjust VRIII doses.
 - For VRIII, a syringe pump is set up with 50 units of human soluble insulin (e.g. Humulin® S or Actrapid®) in 49.5 mL of NaCl 0.9%.
 - The recommended substrate fluid to be administered alongside the VRIII is 0.9% NaCl with 5% glucose and 0.15% KCl (20 mmol/L) at 50 mL/hr. Pure dextrose-containing fluids should be avoided due to the risk of hyponatraemia.
 - During ongoing use of VRIII (>6 hours), serum electrolytes should be checked 4 – 6 hourly to maintain electrolyte balance. If ketoacidosis is suspected, blood ketones should be checked.
 - Additional intravenous fluids may be required as per clinical need, e.g. haemorrhage.
 - Particular care relating to fluid management is needed in women with pre-eclampsia, as they may require fluid restriction alongside intravenous medications. The volume of all administered fluids should be carefully calculated. Provided there are no extra needs (e.g. haemorrhage), this should be kept to 80 mL/h.
- **For elective caesarean section** (Nil by mouth from midnight), book first on the list in the morning. Check blood glucose at 07.00. Omit rapid-acting insulin with breakfast. Omit oral hypoglycaemic agents on the morning of the procedure.
 - Long-acting insulin at a pre-pregnant dose to be given. Replace short acting with IV insulin infusion to be established by 07.30 hours.
 - If an afternoon caesarean section is booked, give the usual morning insulin with breakfast and start IV infusion of insulin no later than 11.00.

4.4. Postnatal care

- NICE recommends that babies of mothers with all types of diabetes should be monitored for neonatal hypoglycaemia for at least 24 hours post-delivery.

4.4.1 Women previously not on insulin before pregnancy

- Insulin infusion or injections should be stopped when the placenta is delivered.

4.4.2 Women with Insulin-treated diabetes before pregnancy

- Follow the plan from diabetic team.

4.4.3 Information and follow-up after birth

- For women with GDM, advise fasting plasma glucose at 6-13 weeks after delivery or HbA1c after 13 weeks to diagnose diabetes post-partum—annual follow-up with HbA1c measurement in women who have a negative post-natal test.
- Women with gestational diabetes have a ten times increased risk of developing type 2 diabetes within five years. Women with GDM are eligible for referral to the National Diabetes Prevention Programme (NDPP) regardless of baseline glycaemia (provided that type 2 diabetes has been excluded).
- Women with pre-existing diabetes should be referred to their routine diabetes care arrangements. Women should be reminded of the importance of contraception to reduce the risks of unplanned pregnancy and the need for pre-conception care when planning future pregnancies.
- Women should be informed about the increased risks of gestational diabetes in future pregnancies and should be offered early self-monitoring of blood glucose or an OGTT in future pregnancies.
- Women who are breastfeeding with pre-existing type 2 diabetes can take metformin after birth but should avoid other oral glucose-lowering drugs. Women should continue to avoid medicines for their diabetes complications that were stopped for safety reasons during the pre-conception period or when pregnancy was identified.
- Women with type 1 diabetes should be screened for post-partum thyroiditis with TSH at 3 and 6 months postpartum.

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

- 5.1. This clinical guideline is designed to provide advice, guidance, and direction to the staff while, leaving room for professional judgment and adaptation to fit individual circumstances.
- 5.2 Review of all cases of stillbirth where diabetes is a factor.

6. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

“Hypoglycaemia in adults with diabetes mellitus-Hospital management guidelines”, March 2021.

“NDHT Integrated Care Pathway for the Management of Diabetic Ketoacidosis in Adults” January 2018.

7. REFERENCES

Joint British Diabetes Society (JBDS) for inpatient care. Managing diabetes and hyperglycaemia during labour and birth. JBDS12. February 2022.

NICE NG3. “Diabetes in pregnancy: management from preconception to the postnatal period”; March 2008, updated February 2020; available from www.nice.org.uk/guidance/ng3

NHS England and Wales. Seventh annual report of National Pregnancy in Diabetes (NPID) Audit; October 2023.

8. PUBLICATION DETAILS

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